Annals of Botany **133**: 73–91, 2024
https://doi.org/10.1093/aob/mcad170, available online at www.academic.oup.com/aob

ANNALS OF BOTANY Founded 1887

SPECIAL ISSUE: MACROALGAE IN A CHANGING WORLD

Loss, resilience and recovery of kelp forests in a region of rapid ocean warming

K. A. Krumhansl[1,*], C. M. Brooks[1], J. B. Lowen[1], J. M. O'Brien[1], M. C. Wong[1] and C. DiBacco[1]

[1]*Fisheries and Oceans Canada, Bedford Institute of Oceanography, Dartmouth, Nova Scotia, B2Y 4A2, Canada*
[*]*For correspondence. E-mail: Kira.Krumhansl@dfo-mpo.gc.ca*

Received: 27 June 2023 Returned for revision: 6 September 2023 Editorial decision: 19 October 2023 Accepted: 8 November 2023

• **Background and Aims** Changes in kelp abundances on regional scales have been highly variable over the past half-century owing to strong effects of local and regional drivers. Here, we assess patterns and dominant environmental variables causing spatial and interspecific variability in kelp persistence and resilience to change in Nova Scotia over the past 40 years.
• **Methods** We conducted a survey of macrophyte abundance at 251 sites spanning the Atlantic coast of Nova Scotia from 2019 to 2022. We use this dataset to describe spatial variability in kelp species abundances, compare species occurrences to surveys conducted in 1982 and assess changes in kelp abundance over the past 22 years. We then relate spatial and temporal patterns in abundance and resilience to environmental metrics.
• **Key Results** Our results show losses of sea urchins and the cold-tolerant kelp species *Alaria esculenta*, *Saccorhiza dermatodea* and *Agarum clathratum* in Nova Scotia since 1982 in favour of the more warm-tolerant kelps *Saccharina latissima* and *Laminaria digitata*. Kelp abundances have increased slightly since 2000, and *Saccharina latissima* and *L. digitata* are widely abundant in the region today. The highest kelp cover occurs on wave-exposed shores and at sites where temperatures have remained below thresholds for growth (21 °C) and mortality (23 °C). Moreover, kelp has recovered from turf dominance following losses at some sites during a warm period from 2010 to 2012.
• **Conclusions** Our results indicate that dramatic changes in kelp community composition and a loss of sea urchin herbivory as a dominant driver of change in the system have occurred in Nova Scotia over the past 40 years. However, a broad-scale shift to turf-dominance has not occurred, as predicted, and our results suggest that resilience and persistence are still a feature of kelp forests in the region despite rapid warming over the past several decades.

Key words: Kelp, climate change, *Saccharina latissima*, *Laminaria digitata*, *Alaria esculenta*, *Agarum clathratum*, *Saccorhiza dermatodea*, *Fucus serratus*, turf, growing degree days, wave exposure, temperature.

INTRODUCTION

Kelp forest ecosystems have been in decline in many regions worldwide due to the synergistic effects of multiple stressors, including climate change, overfishing, invasive species and sea urchin grazing (Krumhansl *et al*., 2016; Filbee-Dexter and Wernberg, 2018; Wernberg *et al*., 2019). Kelp forests support high biodiversity and productivity, and provide ecosystem services valued at over $500 billion US dollars annually (Eger *et al*., 2023). Loss of kelp, therefore, has far-reaching consequences for coastal ecosystems and human societies. As such, these ecosystems have become a focus of conservation and restoration globally (Eger *et al*., 2023).

A global study by Krumhansl *et al*. (2016) identified a small average decline in kelp abundances worldwide over the past half-century, but there was high variability in the trajectories of change on regional scales. Further analysis of these data by Wernberg *et al*. (2019) identified that declines were most prominent in studies of >20 years, suggesting that high variability in short-duration time series may have weakened the overall trend of decline. Moreover, the study revealed that kelp forests globally are vastly understudied due to a lack of focused, ongoing monitoring (Krumhansl *et al*., 2016). The variability, uncertainties and limitations of this global study highlighted that major questions remained about the status of one of the world's most important ecosystems. In the following decade, significant effort has been put towards assessing trajectories of kelp forest change on local scales (Smale, 2019). These studies have revealed strong and persistent declines in many cases (Wernberg *et al*., 2016; Feehan *et al*., 2019; Smale, 2019; Coleman *et al*., 2022; Young *et al*., 2022; reviewed by Smale, 2019), but there have also been many studies documenting kelp persistence, increases over historical timescales and even recovery following periods of loss (Bartsch *et al*., 2016; Reed *et al*., 2016; Pfister *et al*., 2018; Teagle and Smale, 2018; Schroeder *et al*., 2020; Mora-Soto *et al*., 2022; reviewed by Smale, 2019). This high regional variability continues to reinforce the importance of local and regional processes driving kelp forest dynamics, and points to the need for studies that assess drivers of change at small spatial scales (10s to 100s km).

Losses of kelp have largely been attributed to the direct and indirect effects of climate change, but also occur due to overfishing, sea urchin grazing, kelp harvesting, coastal development and invasive species (Wernberg *et al*., 2019). Kelps thrive in cool, nutrient-rich waters (Wernberg *et al*., 2019). Gradual ocean warming and acute marine heatwaves can have

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strong and lasting effects on kelp communities if temperatures exceed thermal optima and mortality thresholds for kelp species (Wernberg *et al*., 2016; Smale *et al*., 2019). Areas of the globe that have harboured the most persistent kelp populations are those where the climate has warmed less rapidly and/or cooled (Bolton *et al*., 2012; Mora-Soto *et al*., 2021, 2022). Studies on the associations between environmental conditions and kelp loss have tended to focus on patterns over broad spatial scales (100s to 1000s of km) capturing latitudinal gradients and regional averages (Wernberg *et al*., 2016; Cavanaugh *et al*., 2019; Smale, 2019). However, variability on sub-regional (10s to 100s of km) scales is increasingly being acknowledged as playing a role in promoting kelp persistence, even within regions experiencing overall declines (Young *et al*., 2016; Starko *et al*., 2022).

Kelp forest ecosystems in Nova Scotia have been the focus of intensive study over the past 50 years (Mann, 1972, 1973; Chapman and Johnson, 1990; Scheibling *et al*., 1999; Filbee-Dexter and Scheibling, 2014). Kelp–urchin dynamics were the dominant driver of change in the system from at least the 1970s to the 2000s, with periods of low kelp abundance during this period being associated with high grazing pressure from sea urchins (Moore and Miller, 1983; Filbee-Dexter and Scheibling, 2014). Disease outbreaks in sea urchin populations, occurring on roughly decadal timescales prior to the early 2000s, caused mass mortalities of urchins followed by broad-scale recoveries of kelp (Moore and Miller, 1983; Filbee-Dexter and Scheibling, 2014). Disease outbreaks in sea urchin populations increased in frequency and intensity during the early 2000s, leading to concurrent mass mortalities over multiple subsequent years and ultimately a decline in sea urchin populations in shallow water (Feehan and Scheibling, 2014). Since the decline of sea urchins, spatial and temporal variability in kelp abundances have largely been attributed to interactions between ocean warming and invasive species (Scheibling and Gagnon, 2009; Watanabe *et al*., 2010; Filbee-Dexter *et al*., 2016). Surveys conducted by Watanabe *et al*. (2010) showed that kelp was common in high abundances coast-wide in 2007, and that recovery of kelps had occurred between 2000 and 2007 following invasion-mediated declines in the early 2000s and a loss of sea urchins (Watanabe *et al*., 2010).

In 2014, Filbee-Dexter *et al*. (2016) conducted repeat surveys of a subset of sites surveyed in 2000 and 2007 and dive surveys at three sites spanning the south shore of Nova Scotia that had been sampled repeatedly between the 1950s and 1980s (Supplementary Data Table S1; Filbee-Dexter *et al*., 2016). They reported declines in kelp abundances at these sites on the order of 85–99 % since the earliest survey periods (Filbee-Dexter *et al*., 2016). In 2014, the kelp ecosystems surveyed along the southwestern shore of Nova Scotia (i.e. south shore) had transitioned to dominance by turf-forming algal assemblages, which is a phenomenon that has been documented worldwide (Filbee-Dexter and Wernberg, 2018). Turf dominance is generally considered a degraded state of the kelp forests primarily due to reductions in carbon storage potential and losses of structural habitat (Filbee-Dexter and Wernberg, 2018; Pessarrodona *et al*., 2021), though some studies have shown increases in macroinvertebrate diversity with transitions to turf (Dijkstra *et al*., 2019). Algal turfs colonize open surfaces rapidly and enhance sedimentation rates, both of which reduce the availability of hard substrate for kelp settlement (Connell and Russell, 2010; Burek *et al*., 2018) and generally inhibit recruitment of microscopic life-history stages (O'Brien and Scheibling, 2018a). These strong feedback mechanisms are thought to lock kelp forests in this degraded state (O'Brien *et al*., 2015; O'Brien and Scheibling, 2016; Filbee-Dexter and Wernberg, 2018), and a return to kelp dominance is predicted to be unlikely as climate change is a major driver of turf-shifts (Filbee-Dexter and Werberg, 2018). It was therefore predicted in 2014 that kelp ecosystems would continue to transition to turf dominance coast-wide in Nova Scotia given ongoing regional warming (Hobday and Pecl, 2014).

An extensive re-shuffling of seaweed-dominated communities is one of the predicted consequence of climate change as a result of species-specific responses to changing environmental conditions (Harley *et al*., 2012). Yet, previous assessments of kelp changes through time in Nova Scotia have focused on the response of the kelp community as a whole (Watanabe *et al*., 2010; Filbee-Dexter *et al*., 2016). The possibility that species-specific responses to climate change may be promoting regional kelp persistence has not been considered in this region, as has been shown elsewhere (Kirihara *et al*., 2006; Smale *et al*., 2015; Teagle and Smale, 2018; Schoenrock *et al*., 2019). One of the earliest coast-wide assessments of kelp populations by Moore and Miller (1983) documented luxuriant kelp (here defined as species in the orders Laminariales and Tilopteridales) communities composed of five main species: *Saccharina latissima*, *Laminaria digitata*, *Saccorhiza dermatodea*, *Agarum clathratum* and *Alaria esculenta.* These species all have different thermal tolerances (Jueterbock *et al*., 2014; Simonson *et al*., 2015) and are likely to have fared differently under warming conditions in Nova Scotia over the past several decades.

The purpose of this study was to assess variability in the persistence of kelp in Nova Scotia on a regional scale. This included three main parts: (1) tracking shifts in the occurrence of individual kelp species at coastal sites surveyed intermittently over the past 40 years, (2) assessing changes in total kelp abundances at sites surveyed repeatedly over the past 22 years and (3) documenting the abundances of kelp species coast-wide today. The second objective of this study is to identify environmental conditions on a site scale (1 km^2) that promote kelp persistence in Nova Scotia. In particular, we aimed to assess whether individual kelp species respond differently to environmental conditions, and if this corresponds to variation in kelp species dominance across coastal sites and recovery from disturbance. This study contributes to our global understanding of how kelp forests are faring under climate change, and provides insights into the potential for kelp persistence and resilience in an ocean warming hotspot.

MATERIALS AND METHODS

Study area

The study area considered mainly the Atlantic coast of Nova Scotia, extending from Canso in the northeast part of the province to Argyle in the southwest (Fig. 1; Supplementary Data Fig. S1, Table S1). We also surveyed sites in southwest

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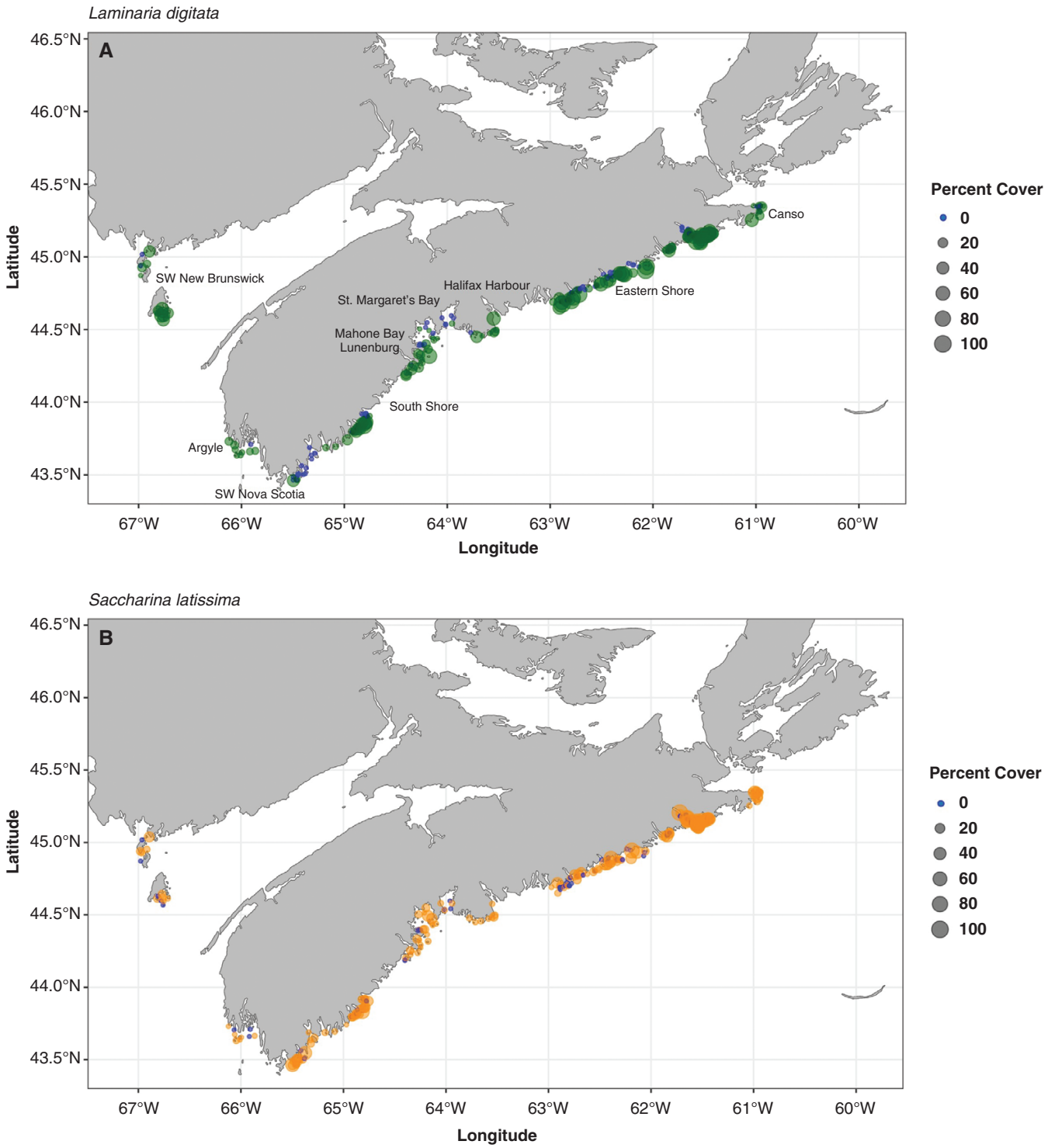

FIG. 1. Percent cover of the two dominant species of kelp: (A) *Laminaria digitata* and (B) *Saccharina latissima* in camera surveys conducted from 2019 to 2022. Points are transparent to show overlap of sites and circle size indicates percent cover. Sites where each species was absent are indicated in blue, and site numbers are shown in Supplementary Data Fig. S1.

New Brunswick around Grand Manan, Deer and Campobello Islands. This study domain spans about 500 km of coastline and includes sites from field surveys conducted in 1982 (Moore and Miller, 1983), 2000, 2007 (Watanabe *et al*., 2010) and 2014 (Filbee-Dexter *et al*., 2016). We also surveyed areas that had previously been poorly studied and that spanned regional gradients of environmental conditions (Lowen and DiBacco, 2022). New sites were selected by randomly choosing coordinates in areas with rocky substrate predicted by a regional substrate model (including bedrock, boulders,

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cobbles and gravel) (Greenlaw and Harvey, 2022) within a depth range of 0–15 m.

Camera surveys

Drop camera surveys were conducted in July–September 2019–2022 ($n = 251$, a complete list of site names, coordinates, sampling depths and dates are provided in Supplementary Data Table S1). Two types of surveys were conducted. The first type (Type 1, conducted at 84 sites) was intended to capture patterns of kelp abundances across the dominant depth range over which kelp occurs by surveying kelp along the shallow–mid (0.5–5 m), and mid–deep (5–12 m, sometimes up to 15 m) extents of kelp beds at each site. Twenty still images were collected along one transect (~150 m) running parallel to shore at each depth contour at each site (two transects, 40 images per site). The camera (GoPro HERO 10) was maintained at 2–4 m off bottom, depending on the visibility. This resulted in a somewhat variable field of view, but was necessary given the wide range of visibility conditions experienced across sites. We used percent cover as our metric of abundance, which is relatively insensitive to this variable field of view. The second type of survey (Type 2) was designed primarily to assess between-site differences in macrophyte cover and capture overall distribution patterns. Therefore, this set of surveys covered a larger number of sites (167) and consisted of short 1–2-min camera drifts at sites shallower than 12 m across a range of substrate types. At each site, the camera system (SPOT X Pro Squid containing a GoPro Hero7) was allowed to drift passively with the motion of the survey vessel over an average distance of ~30 m along which video was recorded. Height of the camera off bottom was similarly adjusted for visibility conditions. Still images were later extracted from video approximately every 4–5 m.

Data processing

Images were analysed to assess the presence/absence and percent cover of all kelp species, including *S. latissima*, *L. digitata*, *Saccorhiza dermatodea*, *Alaria esculenta*, *Agarum clathratum* and the fucoid *Fucus serratus*. Turf cover (fine, e.g. *Bonnemaisona hamifera*, *Antithamnion sparsum*, *Ceramium* sp.; and coarsely branched morphologies, e.g. *Polyides rotunda*, *Chondrus crispus*, *Coccotylus truncata*, *Corralina officanalis*, *Phycodrys fimbriata*) was also assessed, as well as the occurrence of bare substrate (mud, sand, gravel, cobble, boulder, bedrock, mixed), and other large macrophytes identifiable in the images (e.g. *Zostera marina*, *Codium fragile)*. A 10 × 10 point grid was positioned over the extent of each photo, and the cover type at each point was recorded using image annotation software (biigle.de or ImageJ). Point annotations were grouped by photo to generate an estimate of percent cover of each macrophyte/substrate type, and then averaged across photos to generate site-level averages of each cover type. Individual images were also merged with position and depth data to assign each photo a tide-corrected depth to mean lower low water. Tide data were obtained for each site at the nearest tide station as recorded by Environment Canada (https://www.tides.gc.ca/en/tides-currents-and-water-levels).

Historical kelp data

Various sources were consulted to obtain historical data on kelp species distribution, abundance and relative composition. Species presence/absence by depth bin were extracted from a technical report describing the survey conducted by Moore and Miller (1983, $n = 80$ sites), and each species was assigned a semi-quantitative abundance score based on the number of depth bins in which the species occurred during the survey. Data on percent cover of kelps (mean total cover across all species per site) were obtained from Watanabe *et al*. (2010: appendix 4), which included only a subset of the data collected during coast-wide surveys in 2000 and 2007. Sites originally sampled by Watanabe *et al*. (2010) within St. Margaret's, Mahone and Lunenberg Bays were re-sampled by Filbee-Dexter and colleagues in 2014 (Filbee-Dexter *et al*., 2016) and included in our analysis. Sites were matched between surveys, resulting in 16 sites from the 1982 survey re-sampled by the current study, and 19 sites surveyed each in 2000, 2007, 2014 and 2022.

Environmental data

Metrics characterizing temperature effects on key life history processes for the two dominant species of kelp, *S. latissima* and *L. digitata*, were generated by first extracting data from the literature on physiological temperature thresholds, and full and optimal temperature ranges for each species and life history process. These include sporophyte growth (optimal and full ranges), mortality (threshold), sporogenesis (optimal range), gametogenesis (optimal range), gametophyte growth (optimal range) and gametophyte mortality (threshold) (Table 1). Temperature-life history data were collected by searching Web of Science databases using these species names, search terms for the life history processes described above, and '*temperature', '*range' and '*threshold'. Studies from the western Atlantic (Nova Scotia, Gulf of Maine) were prioritized, where possible, but studies from the eastern Atlantic were also included. This search yielded results for each species/life history combination (Table 1).

These thresholds and temperature ranges were then used to calculate a variety of metrics, including the number of days above thresholds (sporophyte growth maximum), and growing degree days for temperature thresholds and ranges. Growing degree days (GDD) is a measure of heat accumulation and is calculated using the following equation as in Lowen and DiBacco (2022):

$$GDD = \sum_{start\ \ date}^{end\ \ date} Ti - TTh$$

where Ti represents daily sea surface temperature (SST) and TTh is the temperature threshold for the life history process represented (e.g. upper threshold for sporophyte growth). Where ranges are specified (e.g. 10–15 °C for sporophyte growth) GDD is calculated when $Tl < Ti < Tu$, with Tl representing the lower temperature threshold in the range and Tu representing the upper threshold in the range. We also calculated annual and seasonal means. Temperature metrics were calculated using multi-scale ultra-high-resolution daily SST data (2002–2022) interpolated to 0.01° resolution after collating data from sensors with spatial resolutions that vary from

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TABLE 1. *Temperature ranges and thresholds associated with biological parameters, including sporophyte growth, sporophyte mortality, sporogenesis, gametogenesis, gametophyte growth and gametophyte survival for* Saccharina latissima *and* Laminaria digitata. *Values were obtained from the literature.*

Species	Biological parameter	Temperature (°C)	Reference
Saccharina latissima	Sporophyte full growth window	5–21	Bolton and Luning, 1982; Simonson *et al*., 2015
	Sporophyte optimal growth window	10–15	Bolton and Luning, 1982; Simonson *et al*., 2015
	Sporophyte upper growth threshold	21	Bolton and Luning, 1982; Simonson *et al*., 2015
	Optimal window sporogenesis	10–15	Park *et al*., 2017; Luning, 1988
	Optimal window gametogenesis	5–15	Luning, 1980
	Optimal window gametophyte growth	10–19	Luning, 1980
	Survival threshold gametophyte	20	Luning, 1980
Laminaria digitata	Sporophyte full growth window	5–21	Bolton and Luning, 1982; Simonson *et al*., 2015
	Sporophyte optimal growth window	10–15	Bolton and Luning, 1982; Simonson *et al*., 2015
	Sporophyte upper growth threshold	21	Bolton and Luning, 1982; Simonson *et al*., 2015
	Optimal window sporogenesis	5–10	Bartsch *et al*., 2013
	Optimal window gametogenesis	5–15	Martins *et al*., 2017; Luning, 1980
	Optimal window gametophyte growth	15–18	Martins *et al*., 2017; Luning, 1980
	Survival threshold gametophyte	20	Luning, 1980

1 to 8 km (https://podaac.jpl.nasa.gov/dataset/MUR-JPL-L4-GLOB-v4.1). Note, the sporophyte mortality threshold for both *S. latissima* and *L. digitata* has been recorded at 23 °C (Bolton and Luning, 1982; Simonson *et al*., 2015) but the occurrence of temperatures above this threshold in the region over the study period was so rare that this metric was not used in the analysis. Instead, days above the growth optimum were used. We also incorporated a metric of wave exposure in our analysis generated by O'Brien *et al*. (2022). This metric characterizes relative wave exposure by combining weighted fetch measurements with modelled wind speeds and frequency, following Keddy (1982).

Temperature metrics were generated annually and seasonally (winter = January–March, spring = April–June, summer = July–September, autumn = October–December), and were also processed into anomalies over baseline conditions. The baseline period was 1999–2010, representing a period of stable conditions prior to a shift to a warmer period from 2011 to the present (Brinkman *et al*., 2018) (Fig. 2). Exploratory analyses (linear correlations) identified that changes in kelp abundances between 2014 and 2022 were more strongly correlated with 5-year averaged values (annual and seasonal) rather than values or anomalies calculated for 1- or 10-year periods. Preliminary analyses also identified that percent cover in the contemporary surveys (2019–2022) was more strongly correlated with 5-year averaged annual anomalies over baseline rather than annual and seasonal values and anomalies from 1- and 10-year periods. Only the set of metrics (i.e. 5-year averages or anomalies) that produced the strongest correlations in these preliminary analyses were used in further statistical analyses, as described below.

Data analysis

To assess changes in kelp species distribution and composition between 1982 and 2019–2022, species occurrences for the five kelp species listed above were mapped for each site sampled at each interval (not all were paired between surveys) for visual comparison. Sites sampled in both the 1982 and 2019–2022 surveys were then paired and changes in species occurrences were assessed as: (1) no change – species absent at both intervals, (2) no change – species present at both intervals, (3) species gain between intervals and (4) or species loss between intervals. The depth distributions of species were also compared between intervals by plotting the frequency of occurrence of each species in 2-m depth bins from 0 to 12 m in 1982 and then 2022. Changes in the abundance of all kelp at sites paired between 2000, 2007, 2014 and 2022 were assessed by comparing mean percent cover of total kelp at each site between each interval and from the earliest to the most recent sampling intervals (i.e. 2000–2007, 2007–2014, 2014–2022, 2000–2022) using one-way ANOVA with time interval as a fixed factor using the aov function in R. Pairwise differences were assessed with the TukeyHSD function (stats package in R, v.3.6.2) (R Core Team, 2022). Plots of residuals vs fitted values, histograms of response variables and normal Q–Q plots were assed to verify that our data meet the assumptions of normality and homogeneity of variances. Correlations and statistical significance between the degree of changes in kelp cover and the environmental metrics described above were assessed using simple linear regression for the 2014–2022 period (stats package in R, v.3.6.2) (R Core Team, 2022).

Percent cover in contemporary surveys (2019–2022) was compared statistically between algal species to assess significant differences in species abundances using a one-way ANOVA using the aov function in R (stats package in R, v.3.6.2) (R Core Team, 2022). Assumptions of normality and homogeneity of variances were assessed as above. We then developed generalized linear mixed effects models to assess how environmental conditions influence spatial variability in the abundance of kelp in contemporary surveys (percent cover in 2019–2022). We also included biotic variables in these

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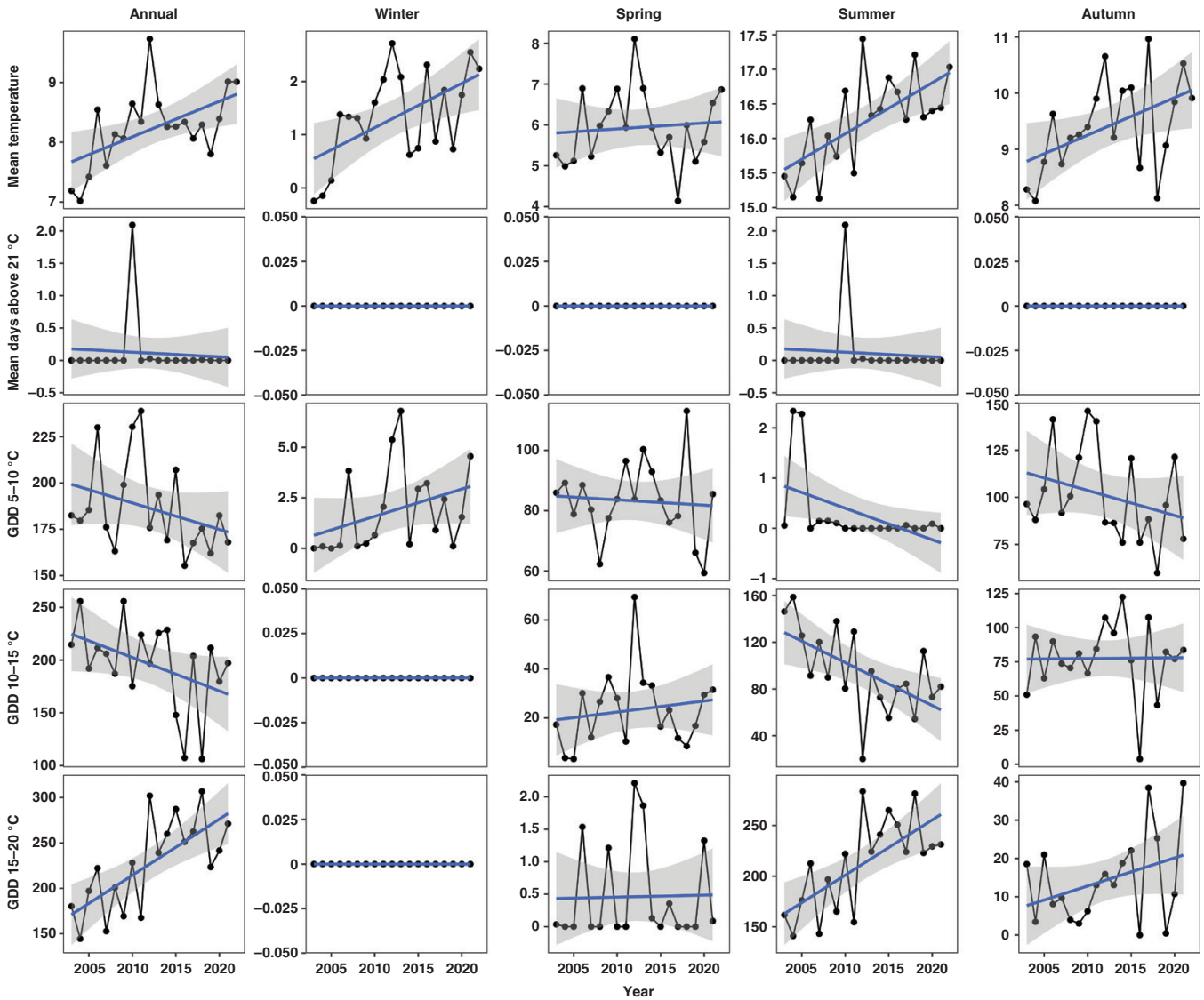

FIG. 2. Trends in temperature metrics through time at sites assessed in contemporary surveys (2019–2022). Shown are the mean number of days above 21 °C, mean temperatures, growing degree day (GDD) in the range of 10–15 °C, GDD in the range of 15–20 °C and GDD in the range of 5–10 °C on annual and seasonal timescales for the years 2003–2022.

models to evaluate the potential for interspecific competition as a source of variation in kelp species abundance. We subsampled sites to those with the lowest (bottom quartile) and highest percent cover (>50 % cover) of each kelp species, as patterns in relation to environmental variables and biological predictors were clearer when we excluded sites that were not clearly dominated by a single habitat type (resulting in sites with kelp dominant vs. little to no kelp). We had a large number of environmental and biological predictor variables, so we elected to use a variable selection approach to assess what metrics in our predictor set best explain variability in kelp abundances. To do so, we generated generalized linear mixed effects models with all combinations of environmental and biological variables and site as a random factor. Since the data are percentages, we used a beta response distribution, which produced a significantly better fit to the data than a gaussian distribution. We compared models with all combinations of variables using the corrected Akaike information criterion (AICc) and cumulative model weights (dredge function in MuMin in R) (Barton, 2023) and selected the top ten models (delta AICc < 2). Model averaging was then performed using AICcmodavg (Mazerolle, 2023) to generate coefficient estimates and relative variable importance rankings (Tredennick *et al.*, 2021). Coefficient estimates were standardized to the mean and standard deviation to also assess the relative effects (i.e. effect size) of selected variables on kelp percent cover.

RESULTS

Environmental conditions

Annual and seasonal mean temperatures at the sites surveyed in 2019–2022 have increased steadily over the past 22 years, with

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a noticeable increase in temperatures between 2003–2010 (Fig. 2). Increases in mean temperatures have been most dramatic in winter, summer and autumn, and weaker in spring (Fig. 2). Temperatures have remained below the 21 °C threshold for *S. latissima* and *L. digitata* growth for the whole 22-year period, with the exception of 2010 (Fig. 2). We plot GDD for three temperature ranges to illustrate general trends in the temperature conditions at temperatures most relevant to kelp: 5–10, 10–15 and 15–20 °C (Fig. 2, Table 1). Spring GDDs have been relatively stable at all three temperature ranges from 2000 to 2022 compared to other seasons, while winter GDD in the range 5–10 °C has increased over time (Fig. 2). In contrast, summer GDD in the 5–10 and 10–15 °C temperature ranges have decreased since 2000, while GDD in the 15–20 °C range has increased strongly (Fig. 2). Autumn GDD has been relatively stable compared to other seasons, but there have been gradual decreases in GDD at 5–10 °C, relative stability at 10–15 °C and increases in the 15–20 °C range (Fig. 2).

Contemporary patterns

The most common cover type on rocky substrates in contemporary surveys was fine turf (19.5 %), followed by coarse turf (14.21 %) and *L. digitata* (14.22 %), then *S. latissima* (9.39 %) and *F. serratus* (8.34 %), and least common were *A. esculenta* (0.05 %), *A. clathratum* (0.09 %) and *S. dermatodea* (0.25 %) ($F_{7, 2592} = 67.88$, $P < 0.001$) (Fig. 3). The abundances of the most common kelp species were highly variable across sites, ranging from 0 to 85 % cover for *S. latissima* and from 0 to 100 % cover for *L. digitata* (Fig. 3), with no strong patterns corresponding to latitude (Figs 1 and 3). There were some sub-regional patterns (10s to 100s of km), with the invasive *F. serratus* mainly occurring in SW Nova Scotia (sites 15–37), Lunenburg and Mahone Bay (71–93), and Canso (240–251), with gaps in this species' distribution between these clusters (Fig. 3). Kelp abundances were also variable on sub-regional scales (10s to 100s of km), with high abundances in SW New Brunswick (sites 1–14), and the south (sites 24–74) and eastern shores of Nova Scotia (sites 113–251) (Figs 1 and 3). Turf communities were dominant at some sites within all subregions, but tended to be the most abundant in the Argyle area of SW Nova Scotia (sites 15–23), and the stretch of coast from St. Margaret's Bay to the approaches to Halifax Harbour (sites 75–112) (Fig. 3).

Historical patterns

Saccharina latissima, *L. digitata, A. esculenta*, *S. dermatodea* and *A. clathratum* were all commonly observed throughout the survey extent by Moore and Miller (1983) (Fig. 4). In 1982, there was a clear zonation of kelp species by depth, with *S. latissima*, *L. digitata* and *A. esculenta* dominant at shallow depths (0–8 m), *S. latissima* and *S. dermatodea* dominating at intermediate depths (4–10 m), and *A. clathratum* and *S. dermatodea* dominating at deep depths (8–12 m) (Fig. 5A). Species were all similarly common, though *S. latissima* was the most common and *A. clathratum* was the least common (Fig. 5B).

Moore and Miller's (1983) results contrasted with recent surveys in this study (2019–2022), during which *S. dermatodea*, *A. clathratum* and *A. esculenta* were very rarely observed (Figs 3B and 4). Zonation patterns by depth were less distinct in contemporary surveys as compared to 1982, with *L. digitata* and *S. latissima* being observed most commonly at relatively shallow depths (0–8 m) (Fig. 5C). In 2019–2022 surveys, *L. digitata* was the most abundant kelp species by percent cover, followed by *S. latissima* (Fig. 3B), but *S. latissima* was still the most common species as measured by occurrence across depth bins (Fig. 5E). The remaining three kelp species had very low percent cover and were not commonly observed across depths in 2019–2022 surveys (Figs 3 and 5C, D). Comparing paired observations at sites surveyed in 1982 and 2019–2022, losses and consistent absences between 1982 and 2019–2022 were most common for *S. dermatodea*, *A. clathratum* and *A. esculenta*, with no gains in these species nor persistence of these species observed at any site (Fig. 5E). In contrast, *S. latissima* and *L. digitata* were gained or remained present between time periods for the majority of sites (Fig. 5E).

Percent cover changes from 2000 to 2022

At the 19 sites re-surveyed in 2000, 2007, 2014 and 2022, we recorded variable changes in total kelp percent cover across time periods as indicated by a significant effect of time period on the change in percent cover ($F_{4, 139} = 12.39$, $P < 0.01$) (Fig. 6). There was a period of kelp increase from 2000 to 2007, with the majority of sites showing positive percent cover changes between these intervals (Fig. 6A). Kelp declined significantly in 2007–2014 relative to the 2000–2007 period (Fig. 6B), and significant increases were then observed in 2014–2022 relative to 2007–2014 ($P < 0.01$) (Fig. 6C). Kelp change over the past 22 years was significantly different from the period of decline in 2007–2014 ($P < 0.01$), but did not differ significantly from the periods of increase of 2000–2007 and 2014–2022 ($P > 0.05$) indicating an overall increase in kelp cover from 2000 to 2022 (Fig. 6D). The degree of change in percent cover of kelp in 2014–2022 was most strongly correlated with GDD in the optimal range for sporophyte growth (10–15 °C) in summer and autumn (Fig. 7). In summer, this variable was negatively correlated with increases in percent cover of kelp, and the trend was opposite in the autumn (Fig. 7).

Environmental and biological metrics predicting spatial abundance patterns in 2019–2022

High spatial variability in *L. digitata* and *S. latissima* percent cover in 2019–2022 surveys is explained by a combination of biological and environmental predictors, with the most important predictors varying between species (Table 2, Fig. 8A,B). For both kelp species, the top ten models were considered to perform equally well in predicting spatial variability in percent cover, as assessed by deltaAICc (Table 2). For both species, the percent cover of turf algae occurred in many or all of the top models and was among the most important predictors of spatial variability in the abundance of both *S. latissima* and *L. digitata* (Table 2, Fig. 8A,B). High turf cover was associated with low kelp percent cover for both species (Table 2, Fig. 8A,B).

For *L. digitata*, wave exposure also ranked among the most important variables and occurred in all ten top models for this

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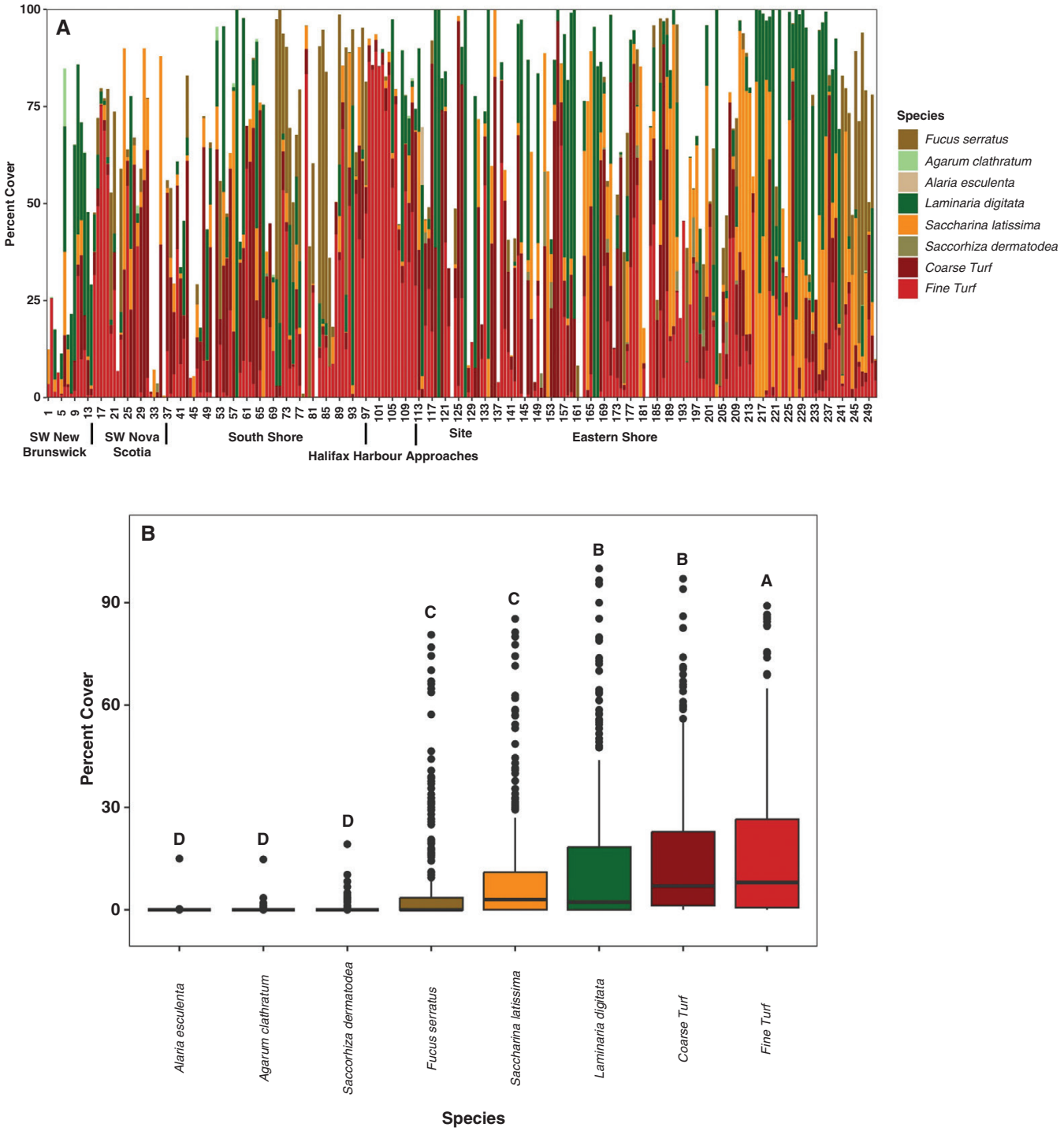

FIG. 3. (A) Percent cover of the dominant macrophyte species/types at all sites surveyed in 2019–2022. Bars are labelled by site number from west to east. (B) Boxplots of abundance of the dominant macrophyte types coast-wide. Letters indicate statistically significant groupings ($\alpha = 0.05$).

species, having the highest scaled coefficient of any predictor considered and a positive relationship with percent cover of this species (Table 2, Fig. 8B). The 5-year average annual anomaly of days above 21 °C was also among the most important predictors, occurring in all of the top models for this species, with sites that had greater anomalies associated with lower percent cover of *L. digitata* (Table 2, Fig. 8B). Five-year average annual GDD anomalies in the 5–10 and 5–21 °C range were also ranked among the most important predictors for *L. digitata* percent cover, having negative and positive effects, respectively (Fig. 8B). Five-year mean temperature, 5-year average annual GDD anomaly of days above 20 °C, *Fucus* spp. percent cover, 5-year average annual GDD anomaly 15–18 °C and 5-year average annual GDD anomaly above 21 °C were also selected

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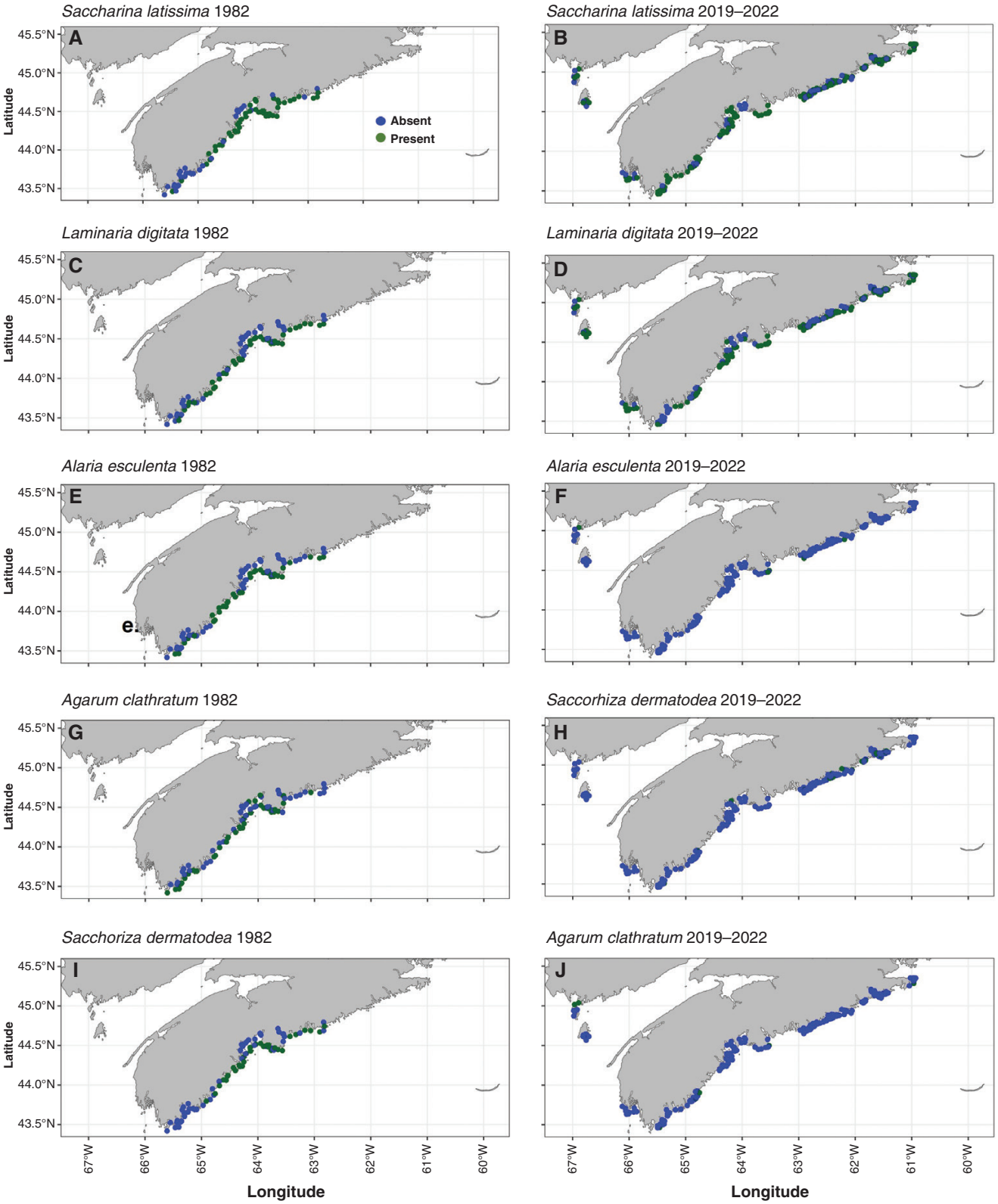

FIG. 4. Presence/absence of the five kelp species in the region in a 1982 survey (A, C, E, G, I) compared to the 2019–2022 surveys (B, D, F, H, J).

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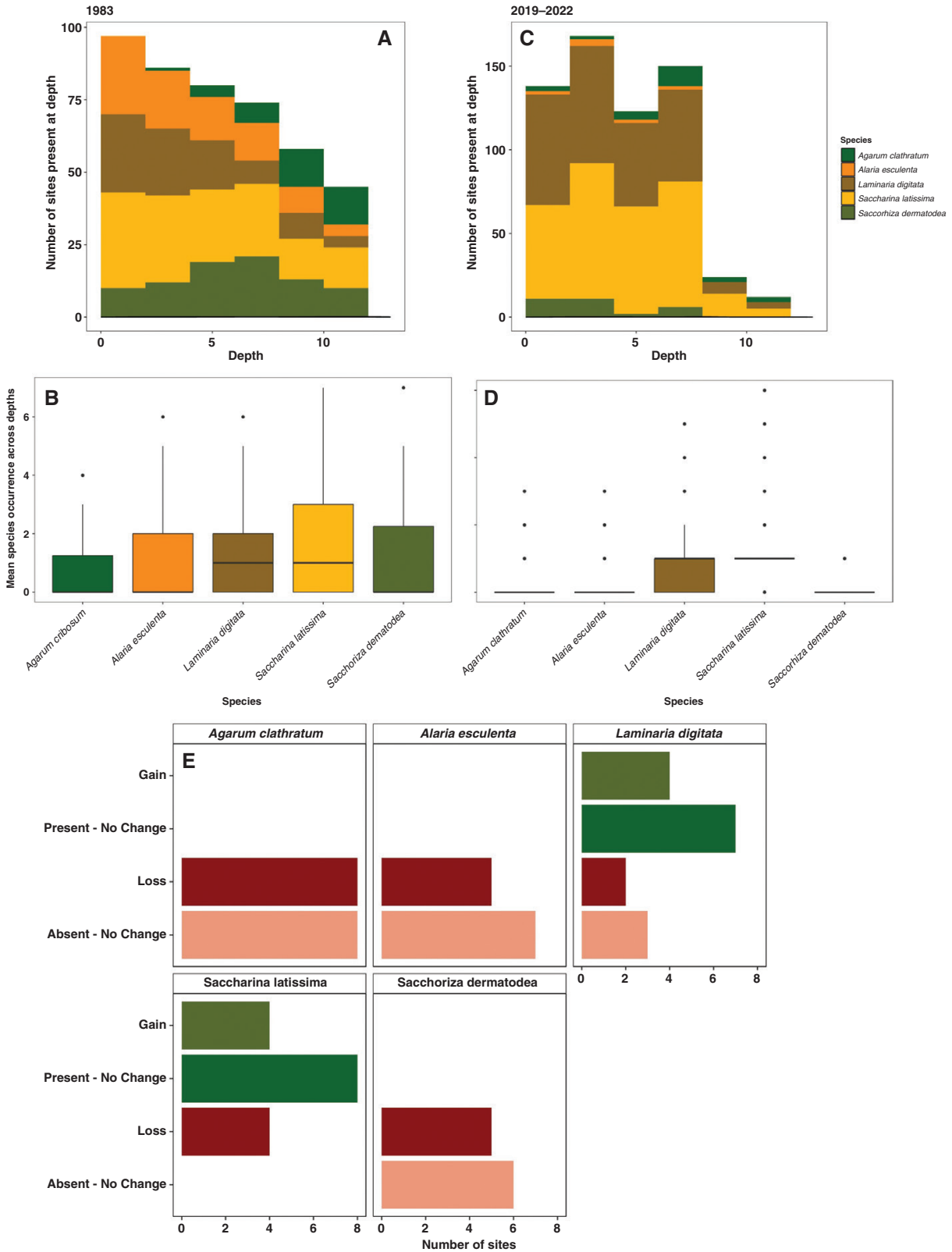

FIG. 5. Depth distributions of kelp species at all sites surveyed in 1982 (A) and 2019–2022 (C). Relative abundances of kelp species in terms of occurrence frequency across depth bins in 1982 (B) and 2019–2022 (D) surveys. Loss, gains or persistent presence/absences of kelp species occurrences at sites sampled in both 1982 and 2019–2022 (E).

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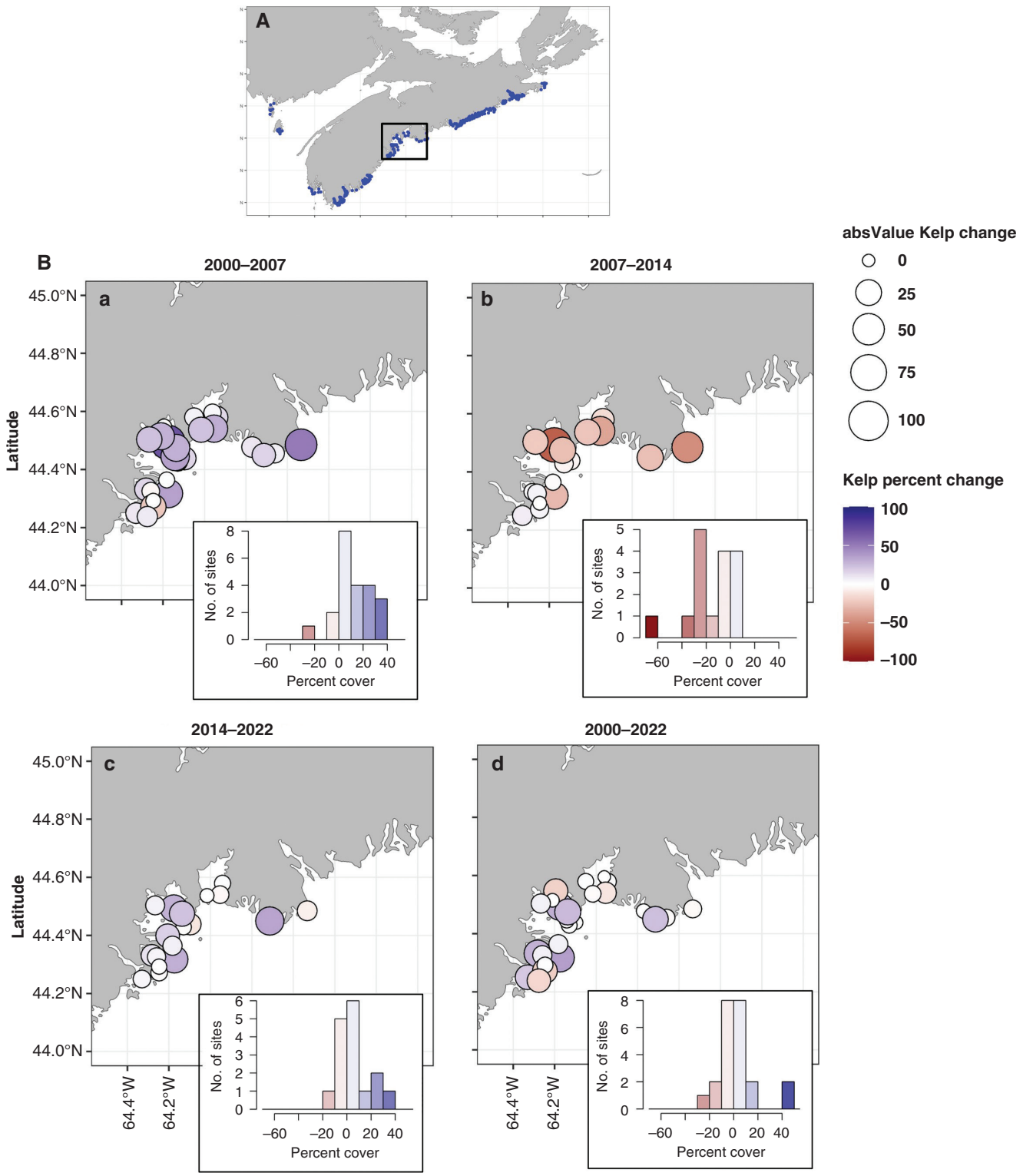

FIG. 6. (A) Site map showing the location of sites resurveyed for percent cover in 2000, 2007, 2014 and 2022. (B) Bubbles show the magnitude (size and colour) and direction of change (colour) between intervals, including (a) 2000–2007, (b) 2007–2014, (c) 2014–2022 and (d) 2000–2022. Inset figures are histograms of the change in percent cover at each site between time intervals.

among the top predictor variables for *L. digitata*, but ranked lower in importance and had relatively small effects on variability in the percent cover of this species (Table 2, Fig. 8B).

The percent cover of *L. digitata* was among the most important predictors of spatial variability in *S. latissima* percent cover, occurring in the models with the highest log-likelihood

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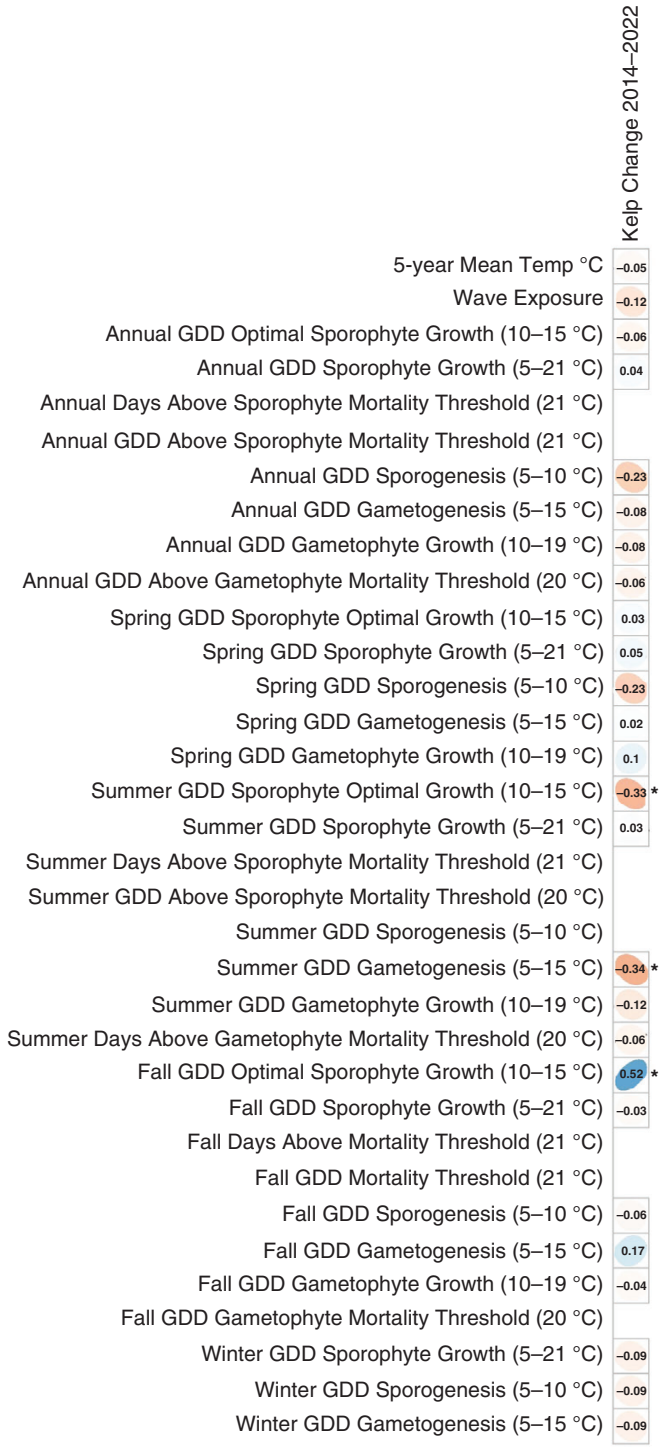

FIG. 7. Correlation plot showing the direction (colour and direction of the ellipse) and magnitude (on of the ellipse, text) of the Pearson's correlation coefficient for all environmental metrics and the change in percent cover of kelp from 2014 to 2022. Note, environmental metrics are values averaged over the past 5 years. Significance is shown with an asterisk at $\alpha = 0.05$.

(Table 2, Fig. 8A). High abundances of *L. digitata* were negatively correlated with abundance of *S. latissima* (Fig. 8A). Wave exposure also occurred in the models with the highest log-likelihood and ranked among the top predictor variables, with a positive effect on percent cover of *S. latissima* (Table 2, Fig. 8A). Five-year mean temperature also had a strong effect on percent cover of *S. latissima*, occurring in seven of the ten top models and ranking moderately high in importance (Table 2, Fig. 8A). High cover of *Saccharina latissima* was associated with lower mean temperatures (Fig. 8A). Annual GDD anomalies above 21 °C also occurred in seven of the top ten models, but ranked relatively low in importance and had only a weak effect on *Saccharina latissima* percent cover (Table 2, Fig. 8A). Annual anomalies in the ranges 5–15, 10–15, 5–21 and 10–19 °C, as well as annual anomaly in the days above 21 °C were also selected in top models for this species, but had relatively small effects and were ranked low in importance (Fig. 8A).

DISCUSSION

Our capacity to predict loss of kelp is contingent on developing an understanding of how kelps respond to climate change, including identifying conditions that lead to kelp loss as well as those that lead to resilience and recovery. The rocky subtidal system in Nova Scotia has undergone dramatic changes over the past 70 years, with periods of large-scale loss due to extensive urchin grazing followed by rapid kelp refoliation after sea urchin mass mortalities (Moore and Miller, 1983; Filbee-Dexter and Scheibling, 2014). In the absence of urchins the rocky subtidal zone has transitioned to a macroalgal-dominated state, but kelp biomass in some areas has declined below historical levels and these sites have undergone a transition to dominance by algal turfs (Filbee-Dexter *et al*., 2016). The present study contributes further to our knowledge of the nature and extent of long-term change in Nova Scotia by showing that the composition of the kelp community has changed dramatically over the past 40 years, but that kelp forests are still a common ecosystem coast-wide. Our results also indicate that kelp forests in Nova Scotia have displayed a persistence and resilience to change over the past couple of decades despite ongoing warming.

Our study is the first to assess species-specific patterns of change in Nova Scotia over historical time periods, and our results show distinct shifts in the composition of the kelp community. We identified a dramatic reduction in the occurrence of three kelp species coast-wide between 1982 and our contemporary period: *A. esculenta*, *A. clathratum* and *S. dermatodea* (Moore and Miller, 1983). In contrast, *S. latissima* and *L. digitata* either persisted at the majority of sites between the two time periods or colonized new sites since 1982. *S. latissima* and *L. digitata* were significantly higher in abundance (percent cover) on average than *A. esculenta*, *A. clathratum* and *S. dermatodea*, and dominate all other kelp species in terms of occurrence across depths down to 12 m in contemporary surveys. This is in contrast to the clear zonation patterns of kelp species with depth observed in 1982, where *S. latissima*, *L. digitata*, and *A. esculenta* dominated at shallow depths (0–8 m), *S. latissima* and *S. dermatodea* dominated at intermediate depths (4–10 m), and *S. latissima* and *A. clathratum* dominated at deeper depths (8–12 m). Moreover, subtidal fucoid species were not noted in the survey conducted by Moore and Miller (1983), whereas the invasive *F. serratus* constituted a significant component of

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TABLE 2. *Top ten generalized mixed effects models predicting percent cover of* Saccharina latissima *and* Laminaria digitata *in surveys conducted from 2019 to 2022, as selected by AICc. Shown are AICc, delta AICc, the Akaike weight and the loglikelihood (LL) of each model.*

Model	AICc	ΔAICc	AICc weight	LL
Saccharina latissima				
GDD anomaly 21 °C + 5-year mean temp	−625.74	0.00	0.18	318.32
GDD anomaly 5–21 °C + GDD anomaly 21 °C	−625.19	0.55	0.14	318.04
GDD anomaly 5–15 °C + GDD anomaly 21 °C + 5-year mean temp	−624.45	1.29	0.10	318.86
GDD anomaly 10–15 °C + GDD anomaly 21 °C + 5-year mean temp	−624.37	1.37	0.09	318.82
GDD anomaly 5–21 °C + anomaly days above 21 °C	−624.19	1.55	0.08	317.54
Laminaria digitata percent cover + Turf percent cover + Wave exposure + 5-year mean temp	−624.17	1.57	0.08	319.95
GDD anomaly 5–21 °C + *Laminaria digitata* percent cover + Turf percent cover + Wave exposure	−624.15	1.60	0.08	319.93
Laminaria digitata percent cover + Wave exposure + 5-year mean temp	−623.88	1.86	0.07	318.58
GDD anomaly 21 °C + Wave exposure + 5-year mean temp	−623.87	1.87	0.07	318.57
GDD anomaly 21 °C + *Laminaria digitata* percent cover + Wave exposure + 5-year mean temp	−623.86	1.88	0.07	319.79
Intercept	−621.94	3.80	0.03	314.14
Laminaria digitata				
GDD anomaly 5–21 °C + GDD anomaly 5–10 °C + Anomaly days above 21 °C + Turf percent cover + Wave exposure	−999.64	0.00	0.18	508.47
GDD anomaly 5–10 °C + Anomaly days above 21 °C + Turf percent cover + Wave exposure + 5-year mean temp	−999.53	0.11	0.17	508.42
Anomaly days above 21 °C + Turf percent cover + Wave exposure + 5-year mean temp	−998.62	1.02	0.11	506.81
GDD anomaly 5–21 °C + Anomaly days above 21 °C + Turf percent cover + Wave exposure	−998.56	1.08	0.11	506.78
GDD anomaly 5–21 °C + GDD anomaly 20 °C + Anomaly days above 21 °C + Wave exposure + Turf percent cover	−998.44	1.20	0.10	507.87
GDD anomaly 5–21 °C + GDD anomaly 5–10 °C + Anomaly days above 21 °C + *Fucus* percent cover + Turf percent cover + Wave exposure	−998.31	1.32	0.09	508.98
CDD anomaly 5–21 °C + GDD anomaly 5–10 °C + Anomaly days above 21 °C + GDD anomaly 20 °C + Turf percent cover + Wave exposure	−997.96	1.68	0.08	508.81
GDD anomaly 20 °C + Anomaly days above 21 °C + Turf percent cover + Wave exposure + 5-year mean temp	−997.77	1.87	0.07	507.54
GDD anomaly 15–18 °C + GDD anomaly 5–21 °C + GDD anomaly 5–10 °C + Anomaly days above 21 °C + Turf percent cover + Wave exposure	−997.44	2.20	0.06	508.55
GDD anomaly 5–10 °C + Anomaly days above 21 °C + GDD anomaly 21 °C + *Fucus* percent cover + Turf percent cover + Wave exposure + 5-year mean temp	−996.07	3.57	0.03	509.05
Intercept	−957.94	41.70	0.00	482.07

the seaweed communities at many sites coast-wide, with an average percent cover (~8 %) that is not significantly different from *S. latissima* (9 %).

Differences in thermal sensitivities between species have probably contributed to the shifts in macroalgal species assemblage composition over the past 40 years. Declines in growth and reproduction and increases in mortality have been observed for *A. esculenta*, *S. dermatodea* and *A. clathratum* at water temperatures >16 °C (Sundene, 1962; Muller *et al*., 2008; Fredersdorf *et al*., 2009; Steinhoff *et al*., 2011; Simonson *et al*., 2015). Mean summer temperatures are now routinely above 16 °C at the sites surveyed in the present study, suggesting that conditions have become less favourable for these species in the region over the past couple of decades (Fig. 2). *A. clathratum* tended to have a deeper distribution than the other kelp species in the 1982 survey, and the species has recently been observed to occur commonly on offshore pinnacles at 20–30 m depth (Peter Lawton, pers. comm.). Bottom temperatures have historically remained consistently colder below ~10 m throughout the year (Saunders and Metaxas, 2007), suggesting that deep water may serve as a cold-water refuge for *A. clathratum.* By contrast, *S. dermatodea* and *A. esculenta* were not observed at 20–30 m during the same dive surveys (Peter Lawton, pers. comm.) and tend to be shallowly distributed elsewhere due to adaptations to high light levels (Wiencke *et al*., 2004; Ronowicz *et al*., 2020). This suggests that light

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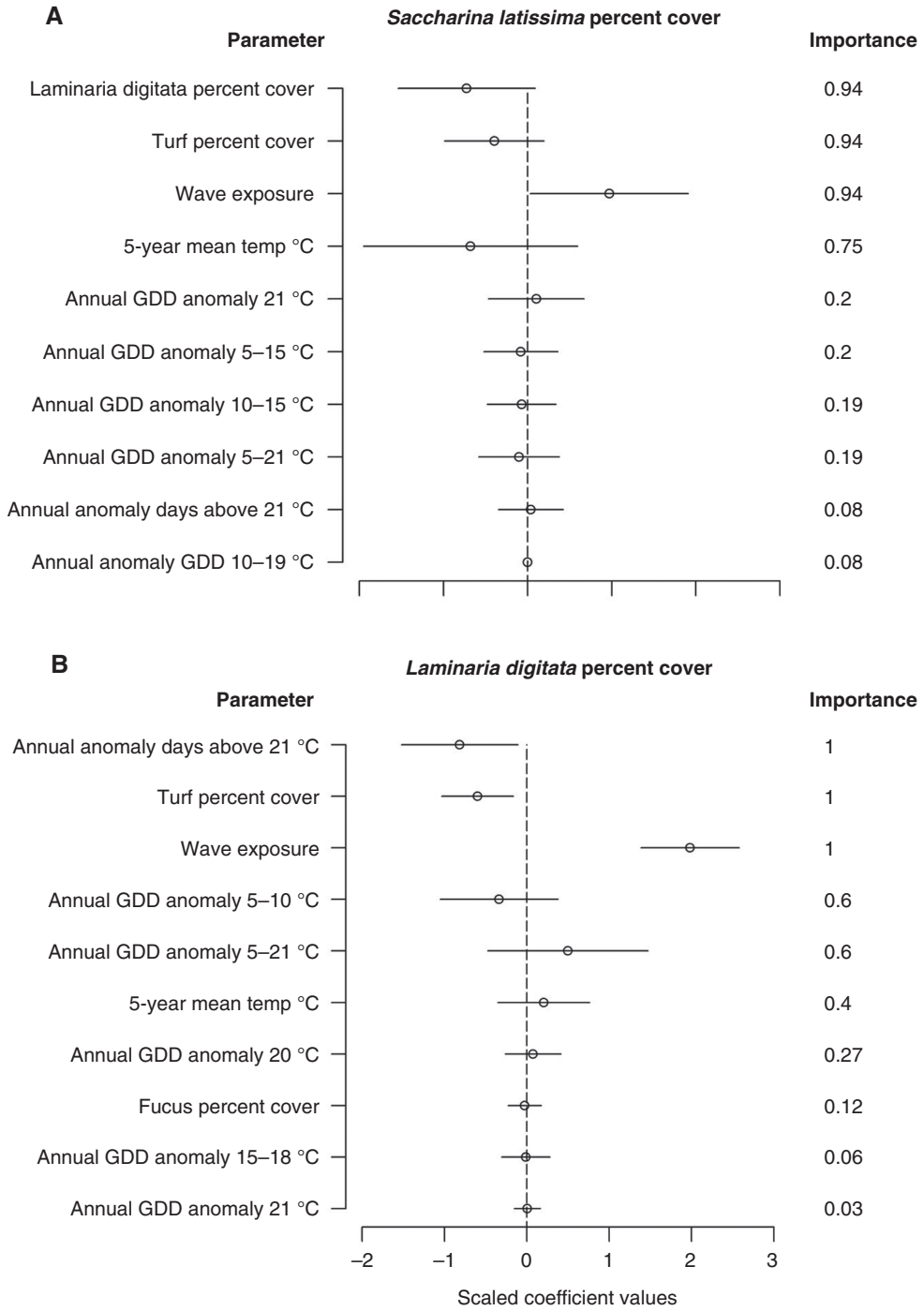

FIG. 8. Scaled coefficient values and relative variable importance as selected by AICc from generalized linear mixed effects models containing environmental and biological variables predicting percent cover of (A) *Saccharina latissima* and (B) *Laminaria digitata.*

limitation may be an important factor constraining shifts to deeper depth distributions in *A. esculenta* and *S. dermatodea*, and may partially explain losses of occurrence of these species. The invasive *F. serratus* has a much higher temperature tolerance than the five native brown algal species described thus far, with optimal growth between 14 and 20 °C (Fortes and Luning, 1980; Figueroa *et al*., 2019), and survival up to 32 °C (Jueterbock *et al*., 2014). We expect that this species will continue to flourish and expand its range throughout Nova Scotia as the climate continues to warm, which will be an important avenue for further study.

Marine species assemblages with differing thermal affinities are known to display diverging responses to sub-regional variation in the magnitude and direction of temperature changes (Kleisner *et al*., 2016). The changes in brown algal species composition occurring in Nova Scotia are similar to observations

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in temperate kelp forests elsewhere globally, with shifts away from cold-tolerant to more warm-tolerant kelp species (Kirihara *et al*., 2006; Smale *et al*., 2015; Teagle and Smale, 2018; Schoenrock *et al*., 2019). The ecological consequences of shifts towards dominance by *L. digitata* and *S. latissima* in Nova Scotia are not known, but community associations vary between kelp species elsewhere, even for kelps with similar habitat characteristics (Tuya *et al*., 2011; Teagle and Smale, 2018; Smale *et al*., 2022). *F. serratus* has a vastly different habitat architecture from the native kelps in Nova Scotia, and therefore probably causes significant changes in associated species communities and ecosystem function, as has been shown for invasive fucoids elsewhere (Wikstron and Kautsky, 2004; Salvaterra *et al*., 2013). An important avenue for further work will be to understand the ecological and ecosystem-service consequences of shifts in the dominant canopy-forming species in Nova Scotia.

Another notable finding of our contemporary surveys was that no urchins were recorded on the Atlantic coast of Nova Scotia in 2019–2022. Small, cryptic urchins have occasionally been observed by divers in the past 5 years (K. Krumhansl, A. Cooper, and J. O'Brien, pers. obs.), but no appreciable accumulations of urchins have been recorded in Nova Scotia since 2015 (Filbee-Dexter, 2016). It is possible that deep refuge populations are being missed from shallow coastal surveys, but dive surveys to 20–30 m on offshore pinnacles in the Eastern Shore also noted a complete absence of sea urchins in these habitats (A. Cooper, pers. comm.). A general absence of urchins at the depths where kelp occurs indicates that grazing dynamics by this herbivore are no longer a dominant driver of the ecosystem. This signals another profound shift in the ecological dynamics of the Nova Scotia system over the past 40 years, and it will be important to focus ongoing research and monitoring effort on the potential for recovery of sea urchin populations on this coast.

Despite these large shifts in species occurrence, kelp communities have shown some capacity to recover from climate-driven disturbances. Regional temperature records show a gradual regional warming trend since the 1990s, with a shift to a warmer period post-2010 (Brinkman *et al*., 2018; Chen *et al*., 2020). Exceptionally high mean and maximum temperatures between 2010 and 2012 (Fig. 2) coincided with the losses of kelp recorded by Filbee-Dexter *et al*. (2016) in 2014 along the south shore of Nova Scotia. These authors attributed kelp loss to the first recorded temperatures in the region >20 °C since the late 1970s (Filbee-Dexter *et al*., 2016). Since 2014, our results show that kelp has increased in abundance on average at the sites monitored by Filbee-Dexter *et al*. (2016) despite regional temperatures remaining higher on average than the pre-2010 period. Our analysis identified that the most important variable explaining spatial variability in the recovery of kelp to these warming events is a positive effect of fall GDD in the range of 10–15 °C. This range corresponds with the optimal temperature range for sporophyte growth for both species (Table 1). The autumn period is a critical time for epibiont-driven losses in Nova Scotia (Krumhansl *et al*., 2011; Krumhansl and Scheibling, 2011a), which can lead to mortality and transitions to turf (O'Brien *et al*., 2015). Relatively high kelp growth during this period might promote resistance to the damaging effects of epibionts, ameliorating storm-driven losses (Krumhansl and Scheibling, 2011b; Krumhansl *et al*., 2011). Moreover, 10–15 °C corresponds to the optimal temperature for sporogenesis in *L. digitata* (Luning, 1988), which occurs in autumn. Higher kelp cover at sites may therefore be attributed in part to more optimal conditions for reproduction at sites with a higher GDD in this range.

We acknowledge that these repeated observations employed to assess percent cover changes between 2000 and 2022 represent only a small number of sites covering a limited spatial extent. However, the presence of kelp populations at high abundances coast-wide and a slight overall increase in kelp abundance over the past 22 years at the sites assessed suggest that the system has a capacity for resilience. Decreases in summer GDD in the ranges of 5–10 and 10–15 °C since 2000 have coincided with increases in GDD in the range of 15–20 °C (Fig. 2). However, temperatures only rose above 21 °C in one year over the past two decades (2010). This means that temperatures have generally stayed below the maximum temperatures for growth (21 °C) and survival (23 °C) for both *S. latissima* and *L. digitata* with the exception of 2 days during 2010. Maintenance of temperatures below these thresholds probably explains how kelp was able to recover from the unusually high temperatures observed between 2010 and 2012 and have remained relatively persistent on the coast over the past 22 years. Also, recovery and persistence has occurred despite continued increases in hurricane frequency and intensity in the Northwest Atlantic (Balaguru *et al*., 2022), suggesting that storms have not been a major stressor to kelp ecosystems in the region over the past 22 years. Other factors that have contributed to kelp recovery elsewhere include improvements to coastal water quality, reduced sedimentation (Tegner *et al*., 1995; Foster and Schiel, 2010) and management of kelp harvest (Gouraguine *et al*., 2021); however, none of these stressors have been documented or observed to impact kelp ecosystems in Nova Scotia and the degree of non-climate human stressors remains low coast-wide (Murphy *et al*., 2019).

The other notable feature of the recovery we observed is that some sites where kelp returned had less than 5 % kelp cover in 2014 and were dominated by turf algae. Our analysis identified that turf is the dominant species group to occupy substrates following kelp loss; however, our results also indicate some capacity for kelp to recover at sites that have transitioned to turf dominance. There are limited examples from other parts of the world of a recovery from turf to kelp (Christie *et al*., 2019) and strong stabilizing mechanisms have been documented for the turf state (Filbee-Dexter and Wernberg, 2018). However, many turf species die back in the autumn and winter (Moy and Christie, 2012), which may open suitable substrate for kelp during the seasonal period that kelps are undergoing sporogenesis and recruitment (Luning, 1988), the life cycle stages that are most susceptible to negative interactions with turf algae (O'Brien and Scheibling, 2018*b*). Moreover, rolling boulders during autmn and winter storms may expose patches of substrate that allow for kelp recruitment. Once recruited, kelp can sweep substrate clean (Toohey *et al*., 2004; Russell, 2007), leading to patch enlargement. We observed a high degree of patchiness of kelp and turf on the scale of 1 m to tens of metres at many of our sites, indicating that these fine-scale biological interactions may be critical for population recovery (O'Brien and Scheibling, 2018*a*) as has been observed elsewhere (Young

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et al., 2016). Our results also suggest caution in using short-term studies to infer long-term stability of alternative ecosystem states (Connell and Sousa, 1983), and indicate that the potential for kelp recovery from turf is an important avenue for ongoing study.

Our analyses indicate that the patchy distribution of kelp on the coast of Nova Scotia at present is driven in part by high spatial variability in environmental conditions, and species-specific responses to this variability. We conducted an analysis to identify the most important environmental determinants of variability in the abundance of the two dominant species of kelp in the system, *S. latissima* and *L. digitata*, in contemporary surveys (2019–2022). Despite having very similar reported temperature tolerances at all life stages (Table 1), our analysis identified that the two species have different sets of environmental controls on their current distributions. Wave exposure was among the top three most important environmental variables predicting spatial variability in percent cover for both species, with higher cover at more wave-exposed locations, particularly for *L. digitata*. It has been suggested that sites with high wave exposure may promote resilience of these two kelp species because of lower temperatures (Filbee-Dexter *et al*., 2016; Witman and Lamb, 2018). However, wave exposure was not strongly correlated to the temperature metrics developed in this study, suggesting another set of mechanisms at play. Attridge *et al*. (2022) attribute higher persistence of kelp populations on exposed shores to a greater capacity to recover from defoliation induced by the bryozoan *Membranipora membranacea*, possibly due to higher nutrients or light levels. Kelps may also be better able to outcompete other seaweeds at higher wave exposures, due to the sweeping action of kelp blades on the substrate that can reduce sediment accumulation and maintain suitable substrate for kelp recruitment (Toohey *et al*., 2004; Russell, 2007).

The metrics of temperature developed in this study had variable effects on the abundance of *S. latissima* and *L. digitata*. *S. latissima* distributions were more strongly related to mean temperatures, with lower cover at sites that were warmer on average over the past 5 years. In contrast, the 5-year mean temperature had a small positive effect on cover of *L. digitata*, but the effect of this variable was weak overall. This suggest a greater sensitivity of *S. latissima* to increases in mean temperature conditions than *L. digitata*. However, 5-year averaged annual anomalies in the number of days above 21 °C was among the most important variables predicting spatial variation in *L. digitata* percent cover, having a negative effect on abundance. This suggests that while *L. digitata* may be less sensitive to rising mean temperatures observed in this study, the areas experiencing warm peaks in temperature are becoming less suitable for this species. Interestingly, metrics characterizing warm events (GDD and number of days above 20 and 21 °C) were not strong predictors of spatial variability in the abundance of *S. latissima*. This is consistent with previous studies showing a greater sensitivity of *L. digitata* to high temperatures than *S. latissima* (Bolton and Luning, 1982; Simonson *et al*., 2015). As temperatures continue to increase and rise more regularly above 20 °C, it is possible that we will see stronger and more persistent losses of *L. digitata* in favour of *F. serratus* and turf assemblages.

Other kelp ecosystems worldwide have displayed a high degree of spatial variability in the resilience and recovery in the face of warming and heatwaves. Cavanaugh *et al*. (2019) and Starko *et al*. (2022) tracked recoveries of *Macrocystis pyrifera* following the 2014–2016 heatwaves along the west coast of North America. Both found a high degree of spatial variability in recovery of kelps that was associated with fine-scale patterns in environmental conditions (Cavanaugh *et al*., 2019; Starko *et al*., 2022). Previous work suggests that the effects of heatwave disturbances and warming have been more extreme at the warm range edge of kelp where they exist closer to their thermal tolerance limits (Wernberg *et al*., 2016; Feehan *et al*., 2019; Filbee-Dexter *et al*., 2020). The high degree of spatial variability we observe in the abundance and resilience patterns of kelp is more consistent with what has been observed in kelp forests in the middle of their range elsewhere (Reed *et al*., 2016; Cavanaugh *et al*., 2019; Schroeder *et al*., 2020). Moreover, the maintenance of regional temperatures within physiological tolerance limits for *S. latissima* and *L. digitata* over the past 22 years suggests that environmental conditions in Nova Scotia are still conducive to kelp persistence, and that the system still has some capacity to resist and recover from future disturbances. Our findings may also foreshadow changes at the cold range edge for kelp in the Arctic, another hotspot of ocean warming where macroalgal habitat is expanding and species composition changes have already been observed (Krause-Jensen *et al*., 2020). There, more temperate species of macroalgae are predicted to gain suitable habitat by the end of the century at the expense of habitat losses for cold-adapted Arctic species (Bringloe *et al*., 2022).

Our results show that a high degree of spatial and temporal variability in kelp abundances and recovery on a sub-regional scale in Nova Scotia is a prominent feature of the system, which points to the importance of monitoring kelp forest change with a high degree of replication at fine spatial and temporal scales. This also points to the importance of site-level variability in environmental conditions for promoting kelp persistence. We identify wave-exposed shores and sites with temperatures remaining below 21 °C as harbouring the most abundant kelp populations, and sites with higher heat accumulation in the range of 10–15 °C in the autumn as the most resilient to disturbance. These populations are likely to be important for the regional persistence of the species as climate disturbances continue to impact more wave-sheltered locations, and should be prioritized for kelp forest conservation. Ongoing research in the region should aim to assess the role of genetics in promoting variability in the response of different kelp lineages to climate change and the potential for kelp species to respond to ongoing changes through evolutionary adaptation. Moreover, it is important to continue monitoring species-specific patterns of change, and further explore the relative roles of environmental and biological interactions on small spatial scales (<1 km) in promoting kelp persistence and resilience to climate disturbances. More broadly, elucidating the features of kelp ecosystems that confer greater resilience will be key to mitigate further losses, promote recovery and maintain the integrity of these foundational habitats across their global range.

SUPPLEMENTARY DATA

Supplementary data are available online at https://academic.oup.com/aob and consist of the following. Fig. S1: Locations

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of sampling sites and corresponding site numbers, numbered from west to east. Table S1: Site metadata.

ACKNOWLEDGEMENTS

We acknowledge Marty King, Tana Worcester and the Marine Conservation Targets programme at Fisheries and Oceans Canada for providing financial support to K.A.K. and C.D.B. and guidance on this work. Shawn Roach, Adam Kennedy and the Smak'nis Marine crew provided valuable field support, and Ryan Stanley provided input on the study aims and design. Bob Miller provided important insights into historical conditions and data collection in his 1982 study. We also thank Neo Paulin, Alyssa Palmer-Dixon, Betty Roethlisberger and Fredrica Jacks for conducting image analysis.

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